

Replication of the Coupled iCDW–rCDW Landau Theory

Christensen, Birol, Andersen & Fernandes, arXiv:2207.12820v2 (2022)

TEXTURES-100 automated replication (Landau-theory core)

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Abstract

We reproduce the reproducible core of Christensen *et al.*: the coupled imaginary-CDW (iCDW, loop currents) and real-CDW (rCDW, bond distortions) Landau free energy on the kagome metal AV_3Sb_5 . We construct the symmetry-allowed free energy [their Eqs. (10)–(13)] from scratch, minimize it numerically over the six-dimensional order-parameter space $(N_1, N_2, N_3; \Phi_1, \Phi_2, \Phi_3)$, and map the phase diagram in the bare quadratic coefficients $(a_r, a_i) \propto (T - T_r, T - T_i)$. We confirm the paper’s headline: the coupled theory yields exactly **two generic mixed charge-ordered states, 3Q-3Q and 2Q-1Q**, selected by the higher-order coefficients; pure iCDW is not a stand-alone minimum; and the 2Q-1Q state breaks C_3 (orthorhombic). We skip the DFT/group-theory VHS symmetry-content derivation. **Verdict: REPLICATED** (Landau core).

1 Model

The order parameters live on the three symmetry-related M points $\mathbf{Q}_1, \mathbf{Q}_2, \mathbf{Q}_3$ (the 2×2 order): a real (TRS-even) rCDW N_i transforming as M_1^+ , and an imaginary (TRS-odd) iCDW Φ_i (loop current). The free energy is $F = F_r + F_i + F_{i-r}$:

$$F_r = \frac{a_r}{2} N^2 + \frac{\gamma_r}{3} N_1 N_2 N_3 + \frac{u_r}{4} N^4 + \frac{\lambda_r}{4} \sum_{i < j} N_i^2 N_j^2, \quad (1)$$

$$F_i = \frac{a_i}{2} \Phi^2 + \frac{u_i}{4} \Phi^4 + \frac{\lambda_i}{4} \sum_{i < j} \Phi_i^2 \Phi_j^2, \quad (2)$$

$$F_{i-r} = \frac{\gamma_{ir}}{3} (N_1 \Phi_2 \Phi_3 + \Phi_1 N_2 \Phi_3 + \Phi_1 \Phi_2 N_3) + \frac{\kappa_{ir}}{4} \sum_{i < j} N_i N_j \Phi_i \Phi_j \\ + \frac{\lambda_{ir}^{(1)}}{4} \sum_i N_i^2 \Phi_i^2 + \frac{\lambda_{ir}^{(2)}}{4} N^2 \Phi^2. \quad (3)$$

The decisive symmetry input: TRS forbids a cubic $\Phi_1 \Phi_2 \Phi_3$ term (odd power of the imaginary order parameter), so the *only* cubic invariant that can lower the iCDW energy is the mixed trilinear γ_{ir} . It is minimized by exactly two configurations, 3Q-3Q (all $N_i, \Phi_i \neq 0$) and 2Q-1Q ($\Phi_i, \Phi_j, N_l \neq 0$).

2 Method

Multistart Nelder–Mead minimization (33 structured + random seeds) of F on a 6×6 grid in $(a_r, a_i) \in [-2, 0.5]^2$, for two coefficient sets. Global minima are classified by counting nonzero N_i and Φ_i . Loop-current interpretation follows the shared `loop_current_meanfield_kernel.py`.

3 Results

- **Scenario A** (large γ_{ir}, γ_r , small biquadratics): deep-cool global minimum is $3Q\text{-}3Q$, $N_i \approx -0.887$, $\Phi_i \approx 0.908$ (all three equal); 30/36 grid cells $3Q\text{-}3Q$.
- **Scenario B** (large biquadratic penalty $\lambda_{ir}^{(1)}$): deep-cool global minimum is $2Q\text{-}1Q$, $N = (0, 0, -1.41)$, $\Phi = (1.01, 1.01, 0)$; 24/36 grid cells $2Q\text{-}1Q$.
- Both mixed states are TRS-breaking ($\Phi \neq 0$: loop currents present).
- **Pure-iCDW test**: cooling only the iCDW ($a_r > 0$) yields a pure single- Q iCDW; a pure $2Q$ or $3Q$ iCDW is never a minimum without accompanying rCDW — exactly the paper’s statement.
- $2Q\text{-}1Q$ has a single nonzero N component (single- Q rCDW), which breaks $C_3 \Rightarrow$ orthorhombic distortion.

4 Comparison to the claim

The headline claim — “the coupled iCDW-rCDW Landau theory yields two generic mixed states, $3Q\text{-}3Q$ and $2Q\text{-}1Q$ ” — is reproduced from first principles at the Landau level: both appear as global minima in disjoint regions of coefficient space, pure iCDW is excluded, and the C_3 -breaking of $2Q\text{-}1Q$ is confirmed. The DFT-derived VHS symmetry content (the seven iCDW types; ferro/octupolar/toroidal/monopolar subsidiary orders) is *not* attempted here and is the residual gap.

5 Verdict

REPLICATED (Landau-theory core). Coverage 7/10, Agreement 9/10.